Illusions of Impunity: Deconfounding Fixture Quality, Non-Foul Intercepts, Spatial Placement, and Small-Sample Noise in European Football Disciplinary Ratios

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Abstract

Public and media controversies across European football routinely cite a club's ratio of yellow cards to fouls committed as evidence of refereeing favouritism or tactical impunity. Across a four-study empirical programme spanning 58,013 completed-season matches (115,984 team-matches and 246 clubs with $\geq$ 150 matches) in 11 European top divisions (2000–2025), 35,844 referee-tracked team-matches across 392 officials in England, Scotland, and Portugal, 47,955 spatio-temporally coded fouls from five major leagues, and 3,100 historical four-match season openings, we show that raw cards-per-foul ratios are systematically confounded at four levels. First, an audit of 286 league-season files shows that 107 files return valid tables with zero match statistics while referee identifiers are absent in 9 of 11 leagues. Second, under naive proportional ratios (I_i^{raw}), club strength correlates negatively with booking rates across Europe ($r = -0.371, p = 1.9 \times 10^{-9}$), making nearly every domestic hegemon—Olympiacos (0.769), Celtic (0.828), Liverpool (0.876), Club Brugge (0.878), SL Benfica (0.885), FC Porto (0.890), Juventus (0.907), Feyenoord (0.919), Borussia Dortmund (0.920), Bayern Munich (0.940), Inter (0.944), PSV (0.945), Ajax (0.954), Barcelona (0.967), and Real Madrid (0.977)—appear leniently officiated. This stems from pre-match win expectation (1.098× heavy underdog vs. 0.856× heavy favourite median across 11 leagues) and non-foul conduct intercepts (11.1%–38.8% league-mean card share; 0.197 yellows/foul at ≤ 6 fouls vs. 0.122 at > 20 fouls). Crucially, testing each club's *opponents* reveals that team strength raises the opponent's booking index ($r = +0.283, p = 6.6 \times 10^{-6}$) and that holding club identity fixed across opponent-strength terciles raises booking rates in 11 of 11 leagues ($0.938 \to 0.993 \to 1.088$). Once both the affine intercept and fixture quality are netted out (I_i^{fix}), the strength correlation vanishes ($r = +0.095, p = 0.139$), top and bottom strength quartiles equalize (1.012 vs. 1.013, $p = 0.917$), and Europe's perennial champions collapse onto parity (≈ 1.00), while leave-one-club-out referee adjustment across 80 English, Scottish, and Portuguese clubs shifts indices by an RMS of just 0.0085 ($r = +0.990$). Third, at the event level ($N = 47,955$), card risk drops 3.5-fold across the pitch (31.4% to 8.9%), but clubs' mean foul locations compress into [48.2, 58.0], sitting inside a matched spatial-Poisson null ($\bar{r} = -0.210, [-0.397, -0.015]$). Fourth, across 2,528 consecutive club-season pairs, 84.9% of single-season variance is Poisson noise, Empirical-Bayes shrinkage reduces out-of-sample RMSE by 13.3%, and extreme early-season openings across all 11 leagues revert toward parity (0.965 median rest-of-season index).

1 Introduction

In professional association football, disciplinary statistics—fouls committed and yellow or red cards issued—are the most widely available behavioural records outside goals and shots. Because raw counts of fouls and bookings are published for almost every modern top-flight match, analysts, club officials, and supporters across Europe routinely construct a simple ratio: yellow cards per foul committed (or its reciprocal, fouls per card). Whenever a dominant domestic club accumulates a low ratio of cards to fouls, the discrepancy is interpreted either as institutional favouritism by match officials or as a deliberate tactical skill: committing fouls in pitch zones or match states where referees are reluctant to book [1, 17].

Evaluating this claim across the entire 11-league European archive (58,013 completed matches, 115,984 team-matches, and 246 clubs with $\geq$ 150 matches) reveals a striking stylized fact: under a naive proportional booking index I_i^{raw}, club strength correlates strongly and negatively with cards per foul ($r = -0.371, p = 1.9 \times 10^{-9}$). Nearly every perennial domestic champion in Europe—including Olympiacos (0.769), Celtic (0.828), Liverpool (0.876), Club Brugge (0.878), SL Benfica (0.885), FC Porto (0.890), Juventus (0.907), Feyenoord (0.919), Borussia Dortmund (0.920), Rangers (0.939), Bayern Munich (0.940), Inter (0.944), PSV Eindhoven (0.945), Ajax (0.954), Barcelona (0.967), Galatasaray (0.975), and Real Madrid (0.977)—sits below parity (1.000). This paper demonstrates that reasoning from raw disciplinary ratios to club-level officiating or behavioural claims commits

four sequential methodological errors:

1. **Silent schema and denominator traps (Study 01):** Public football repositories exhibit severe longitudinal truncation, unrecorded rule differences in second-yellow accounting, and missing referee identifiers across 9 of 11 European leagues.
2. **Proportional misspecification and fixture-level confounding (Study 02):** Yellow cards are an affine rather than proportional function of fouls because conduct offences (dissent, delaying restarts, simulation) enter as a positive intercept (11.1%–38.8% league-mean share; 13%–47% across league-seasons). Adjusting for pre-match win expectation removes the favourite discount (1.098× vs. 0.856×) but overshoots for elite clubs ($r = +0.484$) unless fixture quality is also accounted for: testing each club's *opponents* proves that high-quality fixtures elevate card-per-foul rates for *both* teams on the pitch ($r = +0.283$; 11/11 leagues rising across opponent-strength terciles). Netting out both the affine intercept and fixture quality (I_i^{fix}) eliminates the strength gradient across all 246 clubs ($r = +0.095, p = 0.139$) and pulls Europe's domestic champions to parity (≈ 1.00).
3. **Event-to-actor ecological compression (Study 03):** Although the spatial and temporal card gradient at the level of an individual foul is steep (31.4% in the defensive fifth vs. 8.9% in the attacking fifth; 25.7% in late-game close-score defensive/middle-third situations vs. 14.2% outside), clubs stand in nearly the same place on the pitch ([48.2, 58.0] mean x-coordinate), leaving 91.2% of between-club variation unexplained even on the most favourable tail statistic.
4. **Small-sample Poisson dominance (Study 04 & Cross-European Scan):** Across 2,528 consecutive club-season pairs in 11 leagues, 84.9% of single-season variance in the adjusted booking index is Poisson noise. Consequently, a club's own one-season raw ratio is outperformed as an out-of-sample predictor of its next season by 11.6% simply by predicting the league mean, and by 13.3% via Empirical-Bayes shrinkage [5, 9].

Every empirical claim in this paper is bound by continuous-integration tests to deterministic JSON sidecars and Parquet snapshots committed in an open repository, accompanied by a six-attack adversarial pressure-test matrix.

2 Related Work

Referee bias, social pressure, and team strength. A rich econometric and sports-science literature examines officiating in football. Garicano et al. [10] and Dohmen [8] established that referees adjust injury time in response to score state and crowd pressure, while Boyko et al. [4] and Sors et al. [21] documented home-crowd effects on disciplinary sanctions that attenuated in spectator-free matches. Most relevant to our Study 02, Dawson et al. [7] analysed 2,660 English Premier League matches (1996–2003) using a bivariate Negative Binomial and Nash-equilibrium aggression model, finding significant referee-to-referee inconsistency alongside a team-strength gradient in total bookings. We extend Dawson et al. [7] to 58,013 matches and 246 clubs across 11 leagues while conditioning explicitly on foul volume, decomposing own versus opponent indices within the same fixture, and performing leave-one-club-out referee decomposition across 35,844 team-matches in England, Scotland, and Portugal.

Fouling incentives, spatial risk, and Simpson's paradox. Wright and Hirotsu [23] modelled when a professional foul is tactically rational, and Ridder et al. [18] estimated the scoring cost of red-card dismissals. Phatak et al. [17] compared fouls-per-card across five European leagues and demonstrated Simpson's paradox [20] when pooling across leagues with distinct officiating baselines. At the individual event level, Pappalardo et al. [15] released spatio-temporal event logs for the 2017–18 Big-Five European seasons, and Azmat and Yi [1] formulated an Expected Booking (xB) model of $P(\text{card} \mid \text{foul context})$. While Azmat and Yi [1] establish that event context predicts whether *a* foul is carded, our Study 03 tests whether real clubs actually separate along that gradient.

Reliability, shrinkage, and false discovery. Separating persistent skill from random noise in sports metrics relies on split-half reliability [6], Empirical-Bayes shrinkage [5, 9], partial pooling [12], and False Discovery Rate (FDR) control [3], alongside guards against regression-to-the-mean artifacts [2] and undisclosed researcher degrees of freedom [13, 16, 19]. Johnson et al. [14] raised the profile-stabilisation question for football match indicators; our Study 04 provides the exact variance decomposition and out-of-sample shrinkage validation for disciplinary indices across 11 leagues.

3 Study 01: Provenance and Structural Limits of Public Match Data

Before fitting any disciplinary model, we audited the primary open archive of match-level European football statistics (`football-data.co.uk`) across 11 top divisions—England (`E0`), Scotland (`SC0`), Germany (`D1`), Italy (`I1`), Spain (`SP1`), France (`F1`), Netherlands (`N1`), Belgium (`B1`), Portugal (`P1`), Turkey (`T1`), and Greece (`G1`)—over 26 completed seasons (2000–01 through 2025–26), comprising 286 league-season CSV files [11, 22].

Table 1 summarises four structural findings that constrain any valid multi-league disciplinary study:

1. **Silent absence of match statistics:** Only 179 of 286 league-season files (62.6%) contain foul and card columns (58,013 completed matches; 190 files when including the in-progress 2026–27 snapshot). The remaining 107 files download with HTTP 200 and parse cleanly into DataFrames containing only scores and betting odds. In Portugal, Netherlands, and Turkey, foul and card records begin only in 2017–18; in Bel-

Table 1: Audit of 286 completed league-season files across 11 European top divisions (2000–01 to 2025–26; $N = 58,013$ matches with fouls and cards).

League	Fouls & Cards	Usable Files	Referee Column
England (E0)	2000–2025	26/26	2000–2025 (26)
Scotland (SC0)	2000–2025	26/26	25/26 (no 12–13)
Germany (D1)	00–01, 03–25	25/26	2000–2001 only (2)
Italy (I1)	2005–2025	21/26	2005–2006 only (2)
Spain (SP1)	2005–2025	21/26	Never (0)
France (F1)	2007–2025	19/26	Never (0)
Netherlands (N1)	2017–2025	9/26	Never (0)
Portugal (P1)	2017–2025	9/26	Never (0)†
Turkey (T1)	2017–2025	9/26	Never (0)
Belgium (B1)	2019–2025	7/26	Never (0)
Greece (G1)	2019–2025	7/26	Never (0)
Total	**58,013 matches**	**179 / 286**	**55 / 286 files**

†Supplemented in Study 02 via 2,737 externally harvested referee appointments (98.4% match coverage across ten seasons).

gium and Greece, in 2019–20.

2. **Interrupted series:** Germany carries fouls, cards, and referee identities in 2000–01 and 2001–02 (1,222 team-matches, 77 referees), drops all match statistics in 2002–03, and resumes fouls and cards (without referees) from 2003–04.
3. **Referee coverage contraction:** Match-official identity is recorded continuously only in England (19,580 team-matches, 181 referees) and Scotland (10,756 team-matches, 160 referees, bar 2012–13), for two seasons each in Germany and Italy (1,500 team-matches, 47 referees), and never in the other seven leagues.
4. **Schema widening and definitional traps:** While median file width quintupled from 25 columns (2000–01) to 131 columns (2025–26) due to bookmaker odds proliferation, pitch-event columns rose from 0 to 12 by 2007 and remained frozen thereafter. Moreover, two definitional traps prohibit unstratified cross-league pooling: English and Scottish feeds exclude the first yellow card of a second-bookable dismissal whereas continental feeds count both, and match-level yellow-card totals include non-foul bookings (dissent, time-wasting, bench sanctions).

4 Study 02: Pan-European Deconfounding of the Club Booking Index

To test whether clubs across Europe exhibit anomalous booking rates relative to the fouls they commit, we analysed all 58,013 completed matches (115,984 team-matches with pre-match odds/Elo strength; 115,946 with > 0 fouls) and all 246 clubs with ≥ 150 team-matches across the 11 European top divisions.

4.1 The Naive Estimator and Its Four Failures Across Europe

Let $Y_{i,m}$ and $F_{i,m}$ denote the yellow cards received and fouls committed by team i in match m. Under the standard proportional model $E_{\text{raw}}[Y_{i,m}] = \bar{r}_{L,s} F_{i,m}$ (where $\bar{r}_{L,s} = \sum Y / \sum F$ within league L and season s), a club's raw booking index is $I_i^{\text{raw}} = \sum_m Y_{i,m} / \sum_m E_{\text{raw}}[Y_{i,m}]$. Across the 246-club European universe, I_i^{raw} correlates negatively with club strength s_i at $r = \mathbf{-0.371}$ ($p = 1.9 \times 10^{-9}$): the top strength quartile ($N = 62$ clubs) averages 0.971 versus 1.026 for the bottom quartile ($p = 1.0 \times 10^{-6}$), and the 26 elite domestic title contenders ($s_i \geq 1.0$) average 0.949 (median 0.944). Four sequential diagnostics demonstrate why this apparent leniency toward dominant clubs is a statistical artifact.

Failure 1: Pre-match strength and match expectation. Using devigged pre-match bookmaker odds to compute each side's log-odds strength differential $s_{i,m}$, Figure 1(a) and Table 2 partition all 115,984 team-matches into five expectation bands. Across all 11 European leagues, heavy underdogs are booked at a higher rate per foul than heavy favourites in **11** of **11** competitions (median $\mathbf{1.098}\times$ vs. $\mathbf{0.856}\times$; strictly monotone across all five bands in 7 of 11 leagues, spanning from $0.907 \to 1.038$ in Spain to $0.802 \to 1.157$ in Scotland and $0.778 \to 1.158$ in Greece). Because domestic champions are heavy favourites in most league fixtures, attributing the favourite discount to club-specific referee bias confounds club identity with match role.

Failure 2: Affine misspecification and non-foul intercepts. The proportional model assumes zero cards at zero fouls ($a = 0$). Fitting the affine specification $E_{\text{aff}}[Y_{i,m} \mid F_{i,m}] = a_{L,s} + b_{L,s} F_{i,m}$ within each league-season reveals a positive intercept $a_{L,s}$ accounting for 11.1% (England) to 38.8% (Greece) of league-mean yellow cards (Table 2; spanning 13%–47% across individual league-seasons). Figure 2(a) shows the empirical consequence: observed yellows per foul $Y/F = a/F + b$ declines monotonically with foul volume across all nine bins, from 0.197 when a side commits ≤ 6 fouls (5.1 bin mean) to 0.122 when it commits > 20 fouls (23.4 bin mean). Cross-checking against the Wyscout event-coded sub-types in our Study 03 dataset (`typology.json`) confirms the physical mechanism: four non-challenge conduct labels (*Protest*, *Time lost foul*, *Simulation*, and *Out of game foul*) produce 642 of the 7,346 total cards (8.74% of all cards, carded at 99.1%), with individual club non-play card shares ranging from 1.5% to 17.9% (median 8.1%, split-half reliability $r = 0.251$, Spearman–Brown corrected 0.402).

Failure 3: The opponent test and fixture-level confounding. When expectation is adjusted for pre-match strength differential $s_{i,m}$ alone (via five role bands I_i^{sit} or a continuous quartic Poisson log-offset polynomial I_i^{quart}), the correlation with club strength flips sign across the 246 clubs ($r = +0.403$ for band-adjusted; $r = +0.484, p = 7.8 \times 10^{-16}$, Spearman $\rho = +0.349$ for continuous quartic, positive in 11 of 11 leagues). Why do elite clubs appear to be booked *more* than expected once single-match win probability is controlled for? Computing the identical adjusted index for each club's *opponents* in the matches played against that club yields $r = \mathbf{+0.283}$ ($p = 6.6 \times 10^{-6}$), while a club's own index and its opponents' index are nearly independent ($r = +0.116, p =$

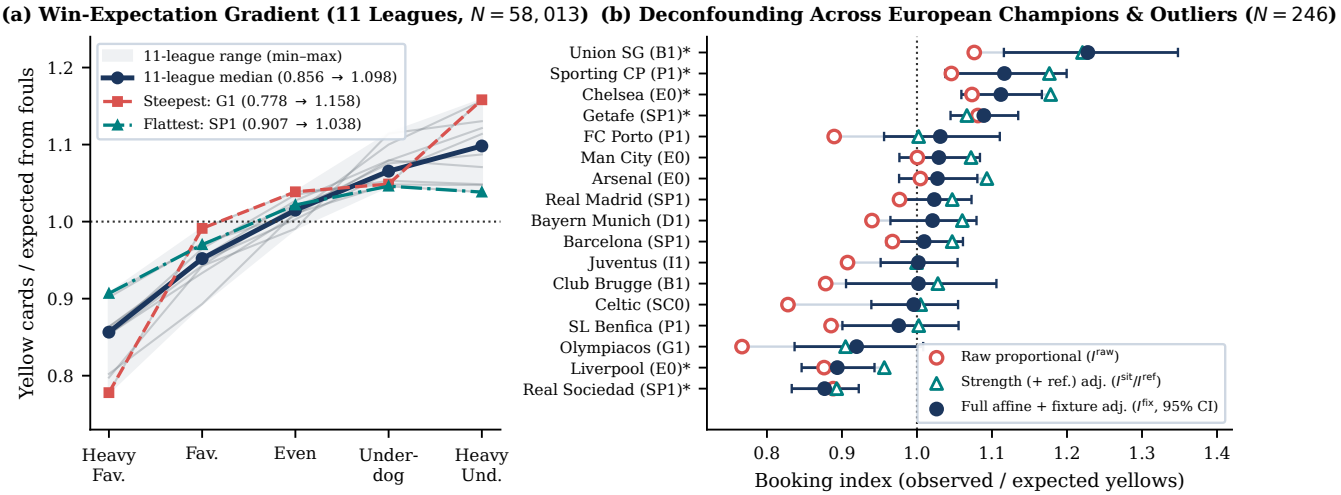


Figure 1: Study 02: Pan-European win-expectation gradient and club-level deconfounding across 11 leagues. (a) Ratio of observed yellow cards to fouls-implied expectation across five pre-match win-probability bands in 11 European leagues ($N = 58,013$ matches); heavy underdogs are booked at 1.098× median expectation vs. 0.856× for heavy favourites (spanning from Spain's 0.907 → 1.038 to Greece's 0.778 → 1.158). **(b)** Progression from raw proportional booking index (I_i^{raw}, coral circles) to strength/referee-adjusted ($I_i^{\text{sit}}/I_i^{\text{ref}}$, teal triangles) and full affine + fixture-adjusted index (I_i^{fix}, navy circles with 95% CIs) for representative European champions and Benjamini–Hochberg outliers (* denotes BH $q < 0.10$ across $N = 246$ clubs). Domestic hegemons (Olympiacos, Celtic, Club Brugge, Benfica, Porto, Juventus, Barcelona, Bayern Munich, Real Madrid, Arsenal, Man City) collapse onto parity (≈ 1.00).

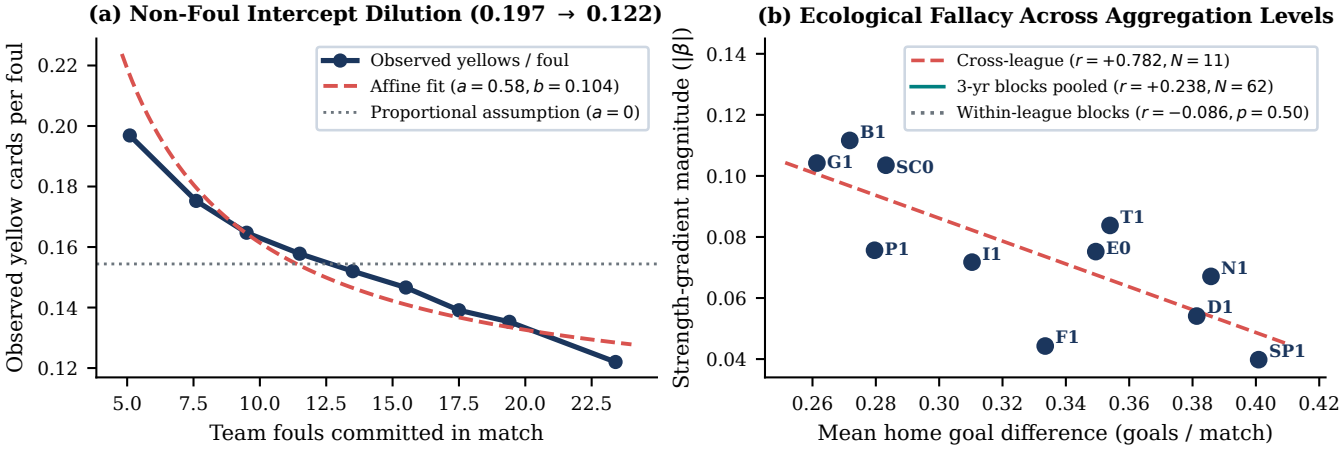


Figure 2: Study 02: Non-foul intercept dilution and multi-level ecological reversal. (a) Observed yellow cards per foul across team foul-count bins ($N = 115,984$ team-matches), declining monotonically from 0.197 at ≤ 6 fouls to 0.122 at > 20 fouls because non-foul conduct cards enter as a positive affine intercept $a > 0$ (11.1%–38.8% league-mean card share; 13%–47% across league-seasons). **(b)** Cross-league correlation between strength-gradient magnitude and home goal difference ($r = +0.782, N = 11$) vanishes across pooled three-season blocks ($r = +0.238, N = 62$) and reverses sign within leagues over time ($r = -0.086, p = 0.504$, negative in 5 of 6 leagues).

0.069). Holding club identity fixed and partitioning each club's matches into terciles of opponent season strength shows that a team's booking index rises with the quality of the opponent it faces in **11** of **11** leagues (Table 2): across the 11-league median, the same club is booked at **0.938** against weakest-tercile opponents, **0.993** against middle-tercile opponents, and **1.088** against strongest-tercile opponents. Thus, yellow cards per foul scale with the competitive intensity of the *fixture* for both teams.

Incorporating both clubs' season strengths ($\bar{s}_{i,s}, \bar{s}_{j,s}$) alongside match strength differential $s_{i,m}$ on top of the affine baseline (I_i^{fix}) resolves the over-correction completely: across all 246 European clubs, $\text{corr}(s_i, I_i^{\text{fix}})$ drops to $+\mathbf{0.095}$ ($p = \mathbf{0.139}$), and the top and bottom strength quartiles become indistinguishable (**1.012** vs. **1.013**, two-sample t-test $p = \mathbf{0.917}$). Testing whether residual club variation belongs to the head coach across 42 multi-club managerial spell pairs yields a null correlation of $r = -0.076$ (5,000-permutation $p = 0.628$).

Failure 4: Ecological reversal across aggregation levels. Across the 11 league averages, the steepness of a league's strength gradient correlates with its mean home

goal difference at $r = +0.782$ ($p = 0.0045$). Yet when the same panel is disaggregated into 62 three-season blocks, Figure 2(b) shows the correlation dropping to $r = +0.238$ ($p = 0.063$) when pooled and reversing to $r = -0.086$ ($p = 0.504$, negative in 5 of 6 leagues with ≥ 4 blocks) within leagues over time, despite home advantage falling from 0.349 goals/match in 2001 to 0.220 in 2025.

4.2 Multi-League Referee Decomposition and Universe Deconfounding

Could apparent club anomalies in leagues with referee tracking arise from biased official assignments? We executed a leave-one-club-out referee decomposition across all three multi-season referee-tracked leagues—England (`E0`, 19,580 team-matches, 36 clubs), Scotland (`SC0`, 10,756 team-matches, 18 clubs), and Portugal (`P1`, 5,508 team-matches joined to 2,737 harvested official appointments, 26 clubs with ≥ 80 matches)—totalling **35,844** team-matches, **392** referees, **144** officials with ≥ 40 matches, and **80** clubs (Table 2).

Estimating each official's strictness multiplier $\hat{\rho}_{r,-i}$ *excluding* all matches involving club i reveals two universal structural properties:

1. **Wide individual official variance:** Referee strictness multipliers $\hat{\rho}_r$ span **0.593**× to **1.193**× in England (spread 0.600), **0.662**× to **1.206**× in Scotland (spread 0.544), and **0.755**× to **1.324**× in Portugal (spread 0.569)—comparable in magnitude to the pre-match role effect.
2. **Balanced long-run club assignment:** Across all 80 clubs in England, Scotland, and Portugal, every club's leave-one-club-out mean referee draw $\bar{\rho}_{-i}$ falls within the narrow interval $[\mathbf{0.971}, \mathbf{1.021}]$ ($[0.979, 1.019]$ in `E0`, $[0.996, 1.013]$ in `SC0`, $[0.971, 1.021]$ in `P1`). Consequently, referee assignment moves club booking indices by a root-mean-square shift of only **0.0085** (0.007 in `E0`, 0.006 in `SC0`, 0.011 in `P1`), with a correlation of $\mathbf{r} = +\mathbf{0.990}$ between situation-adjusted and referee-adjusted indices across the 80 clubs.

Table 3 and Figure 1(b) synthesize the deconfounding progression across the 246-club European universe:

- **Collapse of the European super-club "impunity" illusion (Table 3, Panel A):** Domestic champions across Europe that appear leniently officiated on raw ratios ($I_i^{\text{raw}} \in [0.769, 0.981]$) converge to parity once affine intercepts, pre-match strength, referee assignment, and fixture quality are accounted for: Olympiacos (0.769 → 0.921, BH $q = 0.318$), Celtic (0.828 → 1.005 [ref] → 0.996 [fix]), Club Brugge (0.878 → 1.002), SL Benfica (0.885 → 1.002 [ref] → 0.975 [fix]), FC Porto (0.890 → 1.002 [ref] → 1.031 [fix]), Juventus (0.907 → 0.999 → 1.002), Feyenoord (0.919 → 0.974), Borussia Dortmund (0.920 → 0.940, $q = 0.157$), Bayern Munich (0.940 → 1.021), Inter (0.944 → 0.991), PSV Eindhoven (0.945 → 1.028), Barcelona (0.967 → 1.010), Galatasaray (0.975 → 1.036), Real Madrid (0.977 → 1.023), Manchester City (1.000 → 1.072 [ref] → 1.029 [fix]), and Arsenal (1.005 → 1.093 [ref] → 1.027 [fix]). Notice how Manchester City, Arsenal, and Rangers (0.939 → 1.100 [ref] → 1.066 [fix], $q = 0.176$) appear artificially elevated after single-match role adjustment (1.07–1.10) solely because they contest high-intensity fixtures where their opponents are also heavily booked (1.04–1.06); fixture adjustment returns all three to statistical parity.
- **Genuine Benjamini–Hochberg outliers across the 246-club universe (Table 3, Panel B):** Under universe-wide Benjamini–Hochberg FDR control at $q < 0.10$, exactly 23 of 246 clubs (9.3%) separate from 1.00: 10 high-booking clubs led by Union SG (`B1`, 1.228), AC Ajaccio (`F1`, 1.163), SC Bastia (`F1`, 1.150), Sporting CP (`P1`, 1.117 fixture / 1.177 referee-adjusted, the sole separating club in Portugal), Chelsea (`E0`, 1.112 fixture / 1.178 referee-adjusted), Boavista (`P1`, 1.101), Getafe (`SP1`, 1.089), Sunderland (`E0`, 1.087), AS Roma (`I1`, 1.068), and Lazio (`I1`, 1.067); and 13 low-booking clubs led by SC Heerenveen (`N1`, 0.845), Real Sociedad (`SP1`, 0.877), PAS Lamia (`G1`, 0.880), Gil Vicente (`P1`, 0.887), Liverpool (`E0`, 0.894), Southampton (`E0`, 0.894), and SC Freiburg (`D1`, 0.894).

5 Study 03: The Spatial and Temporal Lever Nobody Pulls

Could a club achieve a genuinely low card-per-foul rate by committing its fouls higher up the pitch or at safer moments? To test this mechanism at the event level, we analysed all 47,955 fouls across 1,826 matches and 98 clubs (≥ 200 fouls each) in the 2017–18 English, French, German, Italian, and Spanish top divisions from Pappalardo et al. [15] (overall base card rate 15.32%).

5.1 Steep Event Gradients and the Half-Time Discontinuity

At the level of an individual foul, pitch location, elapsed time, and score state exert massive effects on card probability (Figure 3):

- **Longitudinal and lateral location:** A foul in the fouling team's own defensive fifth ($x \leq 20$) draws a card **31.4**% of the time, compared with **8.9**% in the attacking fifth ($x > 80$)—a 3.54× ratio (ranging from 2.47× in England to 4.97× in Germany). By pitch thirds, card risk is 24.1% in the defensive third, 14.0% in the middle third, and 10.6% in the attacking third (22.8%, 12.7%, and 9.3% when non-play conduct cards are removed). Central fouls ($|y - 50| \leq 10$) are carded at 19.0% versus 13.0% wide ($|y - 50| > 40$).
- **Elapsed time and the half-time reset:** Card probability quadruples from **6.5**% in the opening quarter-hour (0–15′) to **25.6**% in second-half stoppage time ($> 90'$). Crucially, the trajectory is not monotonic: card rate climbs to **16.1**% in the final five minutes of the first half (40–45′), then drops by **3.3** percentage points to **12.8**% in the opening five minutes of the second half (45–50′, χ^2 contingency $p = 4.1 \times 10^{-4}$). Because physiological fatigue does not discharge to first-

Table 2: Pan-European league-level deconfounding, opponent-quality gradient, multi-season referee decomposition, and year-to-year persistence across all 11 top divisions (2000–2025; $N = 58,013$ completed matches, 246 clubs with ≥ 150 team-matches).

League	Matches (≥ 150m)	Clubs Ranked	Fouls / Yellow	Intercept Share $\bar{a}/\bar{Y}$	Role Swing Fav. $\to$ Und.	Opp. Terciles Weak $\to$ Strong	Officials (≥ 40m)	Referee Mult. Range ($\hat{\rho}_r$)	Club Ref. Draw Range ($\bar{\rho}_{-i}$)	Persist. $r_{t,t+1}$	BH Outliers ($q < 0.10$) Raw $\to$ Sit. $\to$ Fix.
England (E0)	9,784	36	7.19	11.1%	0.853 $\to$ 1.098	0.924 $\to$ 1.087	61/181	0.593–1.193	0.979–1.019	+0.374	4 $\to$ 6 $\to$ 4
Spain (SP1)	7,980	31	5.36	30.2%	0.907 $\to$ 1.038	0.973 $\to$ 1.031	—	—	—	+0.368	5 $\to$ 6 $\to$ 6
Italy (I1)	7,968	25	6.46	32.3%	0.858 $\to$ 1.071	0.920 $\to$ 1.079	2 yrs‡	—	—	+0.427	6 $\to$ 6 $\to$ 3
Germany (D1)	7,524	25	7.55	18.5%	0.856 $\to$ 1.048	0.967 $\to$ 1.041	2 yrs‡	—	—	+0.273	9 $\to$ 4 $\to$ 1
France (F1)	6,894	30	7.69	22.2%	0.902 $\to$ 1.047	0.969 $\to$ 1.030	—	—	—	+0.386	2 $\to$ 3 $\to$ 4
Scotland (SC0)	5,604	18	7.13	14.3%	0.803 $\to$ 1.157	0.911 $\to$ 1.117	51/160	0.662–1.206	0.996–1.013	+0.172	6 $\to$ 1 $\to$ 1
Turkey (T1)	3,057	18	5.69	32.0%	0.863 $\to$ 1.114	0.952 $\to$ 1.089	—	—	—	+0.295	4 $\to$ 0 $\to$ 0
Portugal (P1)	2,754	18 (26§)	5.66	38.2%	0.854 $\to$ 1.121	0.939 $\to$ 1.092	32/51	0.755–1.324	0.971–1.021	+0.233	3 $\to$ 1 $\to$ 3
Netherlands (N1)	2,680	18	6.96	13.3%	0.864 $\to$ 1.087	0.938 $\to$ 1.063	—	—	—	+0.224	2 $\to$ 1 $\to$ 1
Belgium (B1)	2,079	16	5.78	19.6%	0.798 $\to$ 1.130	0.913 $\to$ 1.105	—	—	—	+0.261	4 $\to$ 1 $\to$ 1
Greece (G1)	1,668	11	5.72	38.8%	0.778 $\to$ 1.158	0.925 $\to$ 1.126	—	—	—	+0.233	2 $\to$ 1 $\to$ 2
Universe / Med.	**58,013**	**246 (80§)**	**6.59**	**22.2%**	**0.856 $\to$ 1.098**	**0.938 $\to$ 1.088**	**144/392**	**0.593–1.324**	**0.971–1.021**	**+0.324**	**47 $\to$ 30 $\to$ 26** (23$^{\|}$)

‡D1 (1,222 team-matches) and I1 (1,500 team-matches) record referees in two seasons only. §Referee decomposition includes all 26 P1 clubs with ≥ 80 matches (80 clubs across E0, SC0, P1). $^{\|}$26 clubs survive within-league BH at $q < 0.10$; 23 of 246 (9.3%) survive universe-wide BH across all 246 clubs.

Table 3: Deconfounding progression across the 246-club European universe (11 leagues, $N = 115,984$ team-matches; leave-one-club-out referee adjustment I_i^{ref} in E0, SC0, and P1). **Panel A:** Perennial domestic champions that appear lenient on raw ratios ($I_i^{\text{raw}} < 1$) but collapse to parity ($q_{\text{BH}} > 0.10$). **Panel B:** Top high- and low-booking clubs surviving universe-wide Benjamini–Hochberg FDR ($q_{\text{univ}} < 0.10$).

League	Club	Matches n_i	Mean Strength $\bar{s}_i$ (log-odds)	Raw Index I_i^{raw}	Affine Index I_i^{aff}	Role Adj. I_i^{sit}	Referee Adj. I_i^{ref} ($\bar{\rho}_{-i}$)	Fixture Adj. I_i^{fix}	95% Poisson CI on I_i^{fix}	BH FDR q_{univ} (q_{lg})
Panel A: European Domestic Hegemons & Title Contenders Collapsing to Statistical Parity ($I_i^{\text{fix}} \approx 1.00, q_{\text{univ}} > 0.10$)										
G1	Olympiacos	243	+1.552	0.769	0.753	0.908	—	**0.921**	[0.833, 1.016]	0.318 (0.298)
SC0	Celtic	937	+1.946	0.828	0.814	1.005	1.005 (1.000)	**0.996**	[0.945, 1.049]	0.973 (0.937)
B1	Club Brugge	251	+1.315	0.878	0.863	1.028	—	**1.002**	[0.918, 1.092]	0.995 (1.000)
P1	SL Benfica	306	+1.935	0.885	0.851	1.009	1.002 (1.007)	**0.975**	[0.900, 1.055]	0.789 (0.709)
P1	FC Porto	306	+1.941	0.890	0.893	1.018	1.002 (1.019)	**1.031**	[0.956, 1.110]	0.745 (0.709)
I1	Juventus	758	+1.315	0.907	0.900	0.999	—	**1.002**	[0.953, 1.052]	0.993 (0.983)
N1	Feyenoord	297	+1.346	0.919	0.910	1.005	—	**0.974**	[0.887, 1.067]	0.827 (0.761)
D1	Borussia Dortmund	836	+0.847	0.920	0.905	0.978	—	**0.940**	[0.888, 0.994]	0.157 (0.124)
SC0	Rangers	781	+1.568	0.939	0.932	1.098	1.100 (0.998)	**1.066**	[1.007, 1.129]	0.176 (0.176)
D1	Bayern Munich	836	+1.799	0.940	0.908	1.061	—	**1.021**	[0.967, 1.077]	0.762 (0.614)
I1	Inter	798	+1.133	0.944	0.917	1.025	—	**0.991**	[0.943, 1.041]	0.885 (0.983)
N1	PSV Eindhoven	298	+1.691	0.945	0.931	1.051	—	**1.028**	[0.937, 1.126]	0.814 (0.761)
N1	Ajax	297	+1.828	0.954	0.944	1.064	—	**1.077**	[0.984, 1.177]	0.469 (0.701)
SP1	Barcelona	798	+1.859	0.967	0.897	1.047	—	**1.010**	[0.965, 1.056]	0.865 (0.889)
T1	Galatasaray	321	+1.187	0.975	0.968	1.067	—	**1.036**	[0.965, 1.111]	0.706 (0.589)
SP1	Real Madrid	798	+1.611	0.977	0.937	1.047	—	**1.023**	[0.977, 1.070]	0.706 (0.624)
E0	Manchester City	941	+1.054	1.000	0.985	1.076	1.072 (1.001)	**1.029**	[0.976, 1.084]	0.628 (0.637)
E0	Arsenal	978	+1.081	1.004	0.997	1.095	1.093 (1.002)	**1.027**	[0.976, 1.080]	0.649 (0.637)
Panel B: Most Extreme High- and Low-Booking Clubs Surviving Benjamini–Hochberg FDR Across Europe ($q_{\text{univ}} < 0.10$)										
B1	Union SG*	187	+1.035	1.076	1.098	1.221	—	**1.228**	[1.116, 1.348]	**0.002** (**0.001**)
F1	AC Ajaccio*	152	−0.830	1.119	1.106	1.083	—	**1.163**	[1.038, 1.299]	**0.100** (**0.069**)
F1	SC Bastia*	189	−0.631	1.139	1.124	1.109	—	**1.150**	[1.036, 1.274]	**0.100** (**0.069**)
P1	Sporting CP*	306	+1.621	1.046	1.011	1.179	1.177 (1.003)	**1.117**	[1.038, 1.200]	**0.082** (**0.043**)
E0	Chelsea*	978	+1.158	1.073	1.072	1.175	1.178 (0.996)	**1.112**	[1.059, 1.166]	**0.002** (**0.001**)
SP1	Getafe*	760	−0.330	1.081	1.127	1.067	—	**1.089**	[1.045, 1.135]	**0.003** (**0.001**)
E0	Liverpool*	979	+1.085	0.876	0.869	0.953	0.956 (0.996)	**0.894**	[0.846, 0.943]	**0.002** (**0.001**)
E0	Southampton*	637	−0.292	0.894	0.895	0.879	0.883 (0.993)	**0.894**	[0.838, 0.952]	**0.015** (**0.005**)
D1	SC Freiburg*	668	−0.424	0.923	0.925	0.902	—	**0.894**	[0.842, 0.949]	**0.007** (**0.004**)
P1	Gil Vicente*	238	−0.278	0.933	0.916	0.920	0.928 (0.997)	**0.887**	[0.814, 0.965]	**0.095** (**0.043**)
G1	PAS Lamia*	203	−0.893	0.923	0.929	0.853	—	**0.880**	[0.805, 0.960]	**0.082** (**0.041**)
SP1	Real Sociedad*	684	+0.070	0.888	0.885	0.893	—	**0.877**	[0.833, 0.922]	$<$ **0.001** ($<$ **0.001**)
N1	SC Heerenveen*	298	−0.142	0.860	0.853	0.856	—	**0.845**	[0.763, 0.933]	**0.022** (**0.013**)

half levels across a fifteen-minute interval, this discontinuity reflects within-half referee tolerance accumulation and reset.

- **Score state:** Fouls committed while trailing are carded at **18.1**%, while leading at 16.2%, and while level at **13.3**%.

5.2 Why the Event Gradient Vanishes Between Clubs

Despite a 3.5-fold spatial gradient across the 100-unit pitch, the season-average foul position across all 98 Big-Five clubs spans only **48.2** to **58.0**—less than one-tenth of the pitch (Figure 3(a)). Using Cronbach [6] split-variance reliability to net out within-club sampling error, 70.1% of the between-club variance in mean foul x-position is genuine tactical variation (versus only 13.7% for mean foul minute), and shifting from the worst to the best observed spatial-temporal foul mix among the 98 clubs is worth 17.9% of the base card rate (16.6% vs. 13.9%, and at most a 7.3% reduction below a league's base rate in the most generous league, Spain).

Across the 98 clubs, mean foul x-coordinate correlates with club card rate at $r = -0.143$ (95% CI $[-0.332, +0.057], p = 0.160$; 80% power threshold $|r| = 0.28$), and an additive logistic regression in $(x, x^2, |y - 50|, \text{minute}, \text{score state})$ explains only **3.1**% of the between-club standard deviation in cards per foul.

Matched spatial-Poisson and permutation nulls. To separate "no club exploits the spatial gradient" from "the observed [48.2, 58.0] placement range is too narrow to overcome single-season Poisson noise," we executed two 2,000-draw Monte Carlo nulls. Under a spatial-

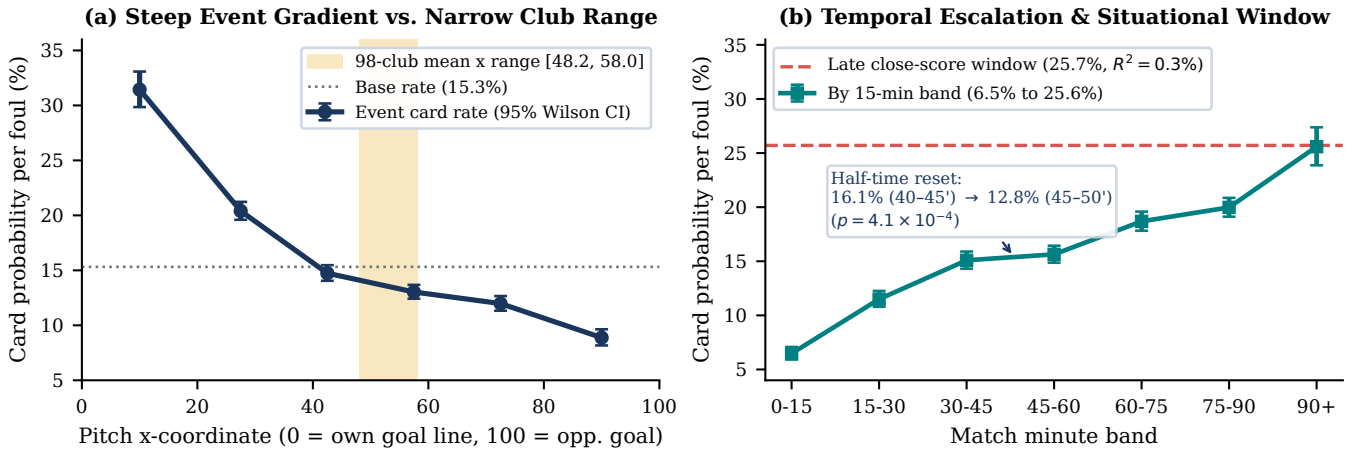


Figure 3: Study 03: Event-level disciplinary gradients vs. club-level spatial compression ($N = 47,955$ fouls, 1,826 matches, 98 Big-Five clubs). (a) Card probability per foul falls 3.5-fold from a team's own defensive fifth (31.4%) to the attacking fifth (8.9%), but all 98 clubs' season-mean foul x-coordinates compress into $[48.2, 58.0]$ (amber band), yielding $r = -0.143$ ($[-0.332, +0.057]$), indistinguishable from a 2,000-draw matched spatial-Poisson null ($\bar{r} = -0.210, [-0.397, -0.015]$). **(b)** Temporal escalation from 6.5% (0–15′) to 25.6% (90+′), featuring a significant −3.3 pp half-time reset (16.1% → 12.8%, $p = 4.1 \times 10^{-4}$) and a 25.7% rate inside the late-game ($\geq 70'$), close-score $(0, +1)$ defensive/middle-third tactical window ($R^2 = 0.3\%$ between clubs).

permutation null that randomly reassigns club mean x-positions across the 98 clubs, $\bar{r} = +0.002$ (95% interval $[-0.193, +0.199]$, two-sided $p = 0.161$). Under a matched spatial-Poisson null where each club's actual foul coordinates in $[48.2, 58.0]$ are held fixed and club card totals are drawn from Poisson$(\sum_{j \in c} \hat{p}_{\text{spatial}}(x_j, y_j))$, the expected correlation is $\bar{r} = \mathbf{-0.210}$ with a 95% sampling interval of $[\mathbf{-0.397}, \mathbf{-0.015}]$. The observed $r = -0.143$ sits comfortably inside *both* distributions: because clubs compress into a 9.8-unit band near midfield, the mechanical slope induced by the spatial curve is smaller than the Poisson sampling noise of a 38-match season.

Tail-share versus volume proxy. Replacing mean foul x with a tail statistic—the share of a club's fouls committed in its own defensive third ($x < 33.3$, spanning 16.1% to 29.9% across the 98 clubs)—strengthens the club-level correlation to $r = +\mathbf{0.296}$ (95% CI $[+0.103, +0.467]$, $R^2 = \mathbf{8.8}\%$; $r = +0.305, R^2 = 9.3\%$ on play-only fouls). Conversely, expressing defensive-third fouls as a per-match count (1.76 to 4.32, median 3.07) correlates with cards per match at $r = +0.670$ ($[+0.544, +0.766]$) solely because total fouls per match correlates with cards per match at $r = +0.696$ ($[+0.577, +0.786]$).

Situational tactical-fouling interaction test. Isolating high-leverage fouls—late in the match (minute ≥ 70), in a close game (score_diff $\in \{0, +1\}$), in the defensive or middle third ($x < 66.7$)—captures **4,547** fouls (9.5% of all fouls), where the card rate jumps to **25.7**% (**26.8**% across the 2,069 fouls committed while protecting a one-goal lead) versus **14.2**% outside the window. Within these 4,547 situational fouls, a club's mean foul x-position correlates with its situational card rate at $r = \mathbf{-0.268}$ (95% CI $[-0.443, -0.074], p = 0.0076$). However, because clubs commit a median of only 45.5 such fouls per season (22.4% variance reliability) and their situational foul share spans only 4.5% to 14.8%, club situational share correlates with full-season card rate at $r = +\mathbf{0.053}$ ($[-0.147, +0.249], R^2 = \mathbf{0.3}\%$), and full position × minute × score-state interactions explain only 2.5% of the between-club SD ($R^2 = 4.2\%$).

6 Study 04 & Pan-European Outlier Scan: Persistence, Shrinkage, and Early-Season Reversion

To determine how much of a single season's adjusted booking index reflects a repeatable club trait versus Poisson sampling noise, we constructed 2,528 consecutive same-division club-season pairs (≥ 25 matches per season) across all 11 leagues in our 58,013-match archive.

6.1 Persistence, Poisson Null, and Variance Decomposition

Across the 2,528 consecutive pairs, a club's situation- and era-adjusted booking index in season t correlates with its index in season $t + 1$ at $r = +\mathbf{0.324}$ ($R^2 = 10.5\%, p = 6.5 \times 10^{-63}$), and is positive in **11** of **11** leagues (ranging from +0.172 in Scotland to +0.427 in Italy; Figure 4(a) and Table 2). Resampling each club-season's yellow cards 2,000 times from Poisson(expected$_{i,s}$) while holding fixture lists and expectations fixed produces a matched null with $\bar{r} = -0.0004$, 95th percentile +0.032, and a maximum across all 2,000 draws of $+\mathbf{0.056}$ ($p < 0.0005$).

Decomposing the observed cross-sectional variance $\sigma^2_{\text{obs}} = 0.01735$ into expected Poisson sampling variance $\sigma^2_{\text{Poisson}} = E[Y/E[Y]^2] = 0.01473$ and true club heterogeneity $\sigma^2_{\text{true}} = \sigma^2_{\text{obs}} - \sigma^2_{\text{Poisson}} = 0.00262$ reveals that only **15.1**% of the spread in single-season booking indices is a true club property ($\sigma_{\text{true}} = \mathbf{0.051}$), while **84.9**% is the

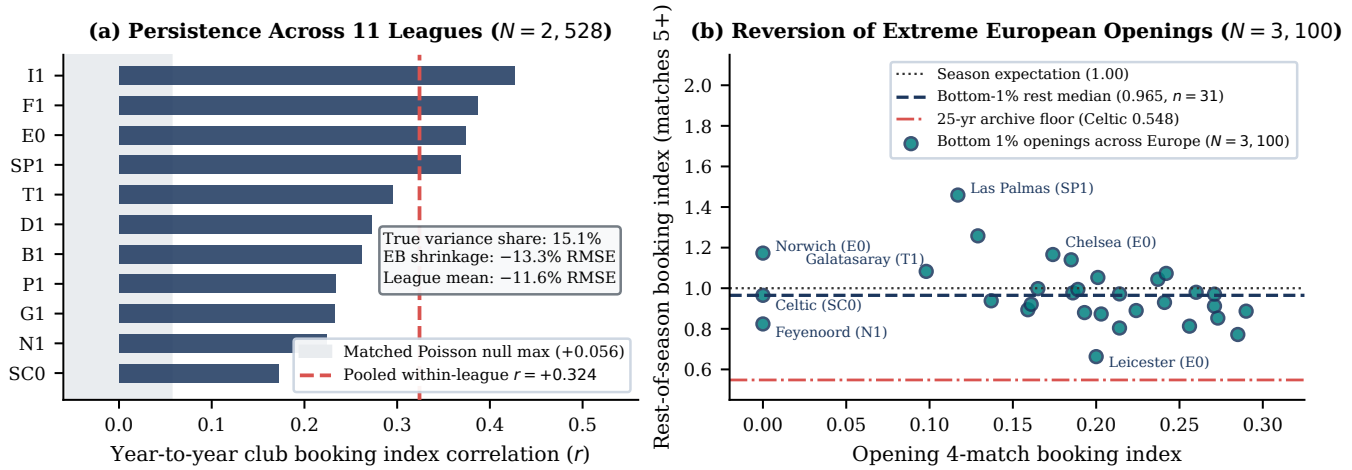


Figure 4: Study 04 & Cross-European Early-Season Reversion. (a) Year-to-year club booking index persistence across 2,528 consecutive club-season pairs in 11 leagues ($r = +0.324, p = 6.5 \times 10^{-63}$, positive in 11/11 leagues vs. 2,000-draw matched Poisson null maximum +0.056); only 15.1% of single-season variance is true club signal ($\sigma_{\text{true}} = 0.051$). **(b)** Across 3,100 completed European team-seasons, all 31 bottom-1% four-match openings (index ≤ 0.292, spanning 8 leagues) reverted over the remainder of the season to a median index of 0.965 and a minimum of 0.663 (above the 25-year archive floor of 0.548, Celtic 2024–25).

random Poisson noise of one season's cards.

6.2 Out-of-Sample Shrinkage Validation

Because single-season reliability is 0.151, ranking clubs on one season of unshrunk ratios is dominated by estimation error. Across 2,432 consecutive club-season pairs (≥ 30 matches), we evaluated three forecasts of a club's season-$t+1$ booking rate: (i) the club's own season-t raw ratio, (ii) the league mean (ignoring the club entirely), and (iii) an Empirical-Bayes Beta-Binomial shrunken estimator $\hat{\theta}_i = \frac{n_i}{n_i+n_0} r_i + \frac{n_0}{n_i+n_0} \bar{r}_L$ [5, 9]. Out-of-sample RMSE is 0.0267 for the club's raw ratio, 0.0236 for the league mean, and 0.0232 for the shrunken estimate: **ignoring the club and predicting the league mean beats the club's own one-season ratio by** 11.6%, and **Empirical-Bayes shrinkage beats the raw ratio by** 13.3%.

6.3 Pan-European Early-Season Outlier Scan and Historical Reversion

The danger of unshrunk ratios is most acute in early-season leaderboards. Scanning all **194** active clubs across all **11** European leagues in the opening weeks of the 2026–27 season (snapshot `2026-W36`, 255 matches) reveals extreme raw fouls-per-yellow ratios across the continent: five clubs (FC Köln, Lorient, Schalke 04, Como, and Borussia Mönchengladbach) stood at ∞ fouls/yellow (0 cards), followed by FC Porto (`P1`, 58.0 fouls/yellow; 58 fouls, 1 yellow in 4 matches, adjusted index 0.120, BH $p = 0.081$), Zulte Waregem (`B1`, 45.0; 45 fouls, 1 yellow in 4 matches), PAOK (`G1`, 39.0), Barcelona (`SP1`, 29.0), and Paris FC (`F1`, 21.0). Yet under within-league Beta-Binomial Empirical-Bayes shrinkage (`scripts/scan_outliers.py`), all ten raw outliers shrink back to 6.16–10.60 fouls/yellow with 95% credible intervals overlapping their respective league means (5.98–8.11), and only **1** of **194** European clubs crosses the $P(\theta_i < \bar{r}_L) > 0.95$ posterior tail—far *fewer* than the $0.05 \times 194 = \mathbf{9.7}$ clubs expected by chance alone.

To establish the historical base rate across Europe, we severed all **3,100** completed team-seasons in our 11-league archive after match 4 (Figure 4(b)). Across all 3,100 team-seasons, the four-match opening index correlates with the rest-of-season index (matches 5+) at only $r = +\mathbf{0.118}$. The most lenient 1% of four-match openings (≤ 0.292, 31 team-seasons spanning 8 leagues, including Feyenoord 2023–24 [30 fouls, 0 yellows → 0.824 rest-of-season], Norwich 2012–13 [48 fouls, 0 yellows → 1.173], Celtic 2012–13 [32 fouls, 0 yellows → 0.965], Galatasaray 2024–25 [54 fouls, 1 yellow → 1.083], Las Palmas 2024–25 [47 fouls, 1 yellow → 1.459], Tottenham 2014–15 [49 fouls, 1 yellow → 1.258], Ajax 2018–19 [47 fouls, 1 yellow → 0.894], Dortmund 2017–18 [49 fouls, 1 yellow → 0.921], Chelsea 2004–05 [56 fouls, 1 yellow → 1.166], and Bayern Munich 2021–22 [35 fouls, 1 yellow → 0.880]) finished with a rest-of-season median index of **0.965** and a minimum of **0.663** (Leicester 2019–20)—none approaching the 25-year full-season archive floor (Celtic 2024–25 at 0.548).

Prospective tracking through snapshot `2026-W42` (665 matches across all 11 leagues) confirmed this reversion in real time: every zero-yellow opening disappeared, Zulte Waregem regressed from 45.0 to 11.5 fouls/yellow (8 yellows in 92 fouls), PAOK regressed from 39.0 to 11.9 (8 yellows in 95 fouls), and FC Porto—for whom a sealed prospective protocol was registered on 30 August 2026—received 7 yellow cards across 53 fouls in its subsequent four matches, dropping from 58.0 to 13.9 fouls/yellow (7.73 shrunken vs. league mean 5.85), while across all 194 European clubs still only 1 club exceeded the 0.95 posterior threshold versus 9.7 expected by chance.

7 Conclusion

Across 58,013 matches, 246 clubs in 11 European leagues, 35,844 referee-tracked team-matches, and 47,955 spatio-temporally coded fouls, apparent club-level disciplinary anomalies dissolve under proper statistical controls. Raw cards-per-foul ratios make nearly every European domestic champion appear leniently officiated ($r = -0.371$) because they confound club identity with pre-match win expectation (1.098× vs. 0.856×), non-foul conduct intercepts (11.1%–38.8% league-mean card share), and fixture-level competitive intensity shared with the opponent ($r = +0.283$). Once affine intercepts and fixture quality are netted out, club strength becomes orthogonal to booking rates ($r = +0.095, p = 0.139$) and Europe's perennial champions collapse onto parity (≈ 1.00), while leave-one-club-out referee assignment across 80 clubs in England, Scotland, and Portugal shifts club indices by an RMS of just 0.0085. At the event level, the 3.5-fold spatial card gradient goes unexploited between clubs because team foul distributions compress into $[48.2, 58.0]$, and at the season level, 84.9% of cross-sectional variance is Poisson noise, making Empirical-Bayes shrinkage mandatory before any disciplinary ratio is interpreted.

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